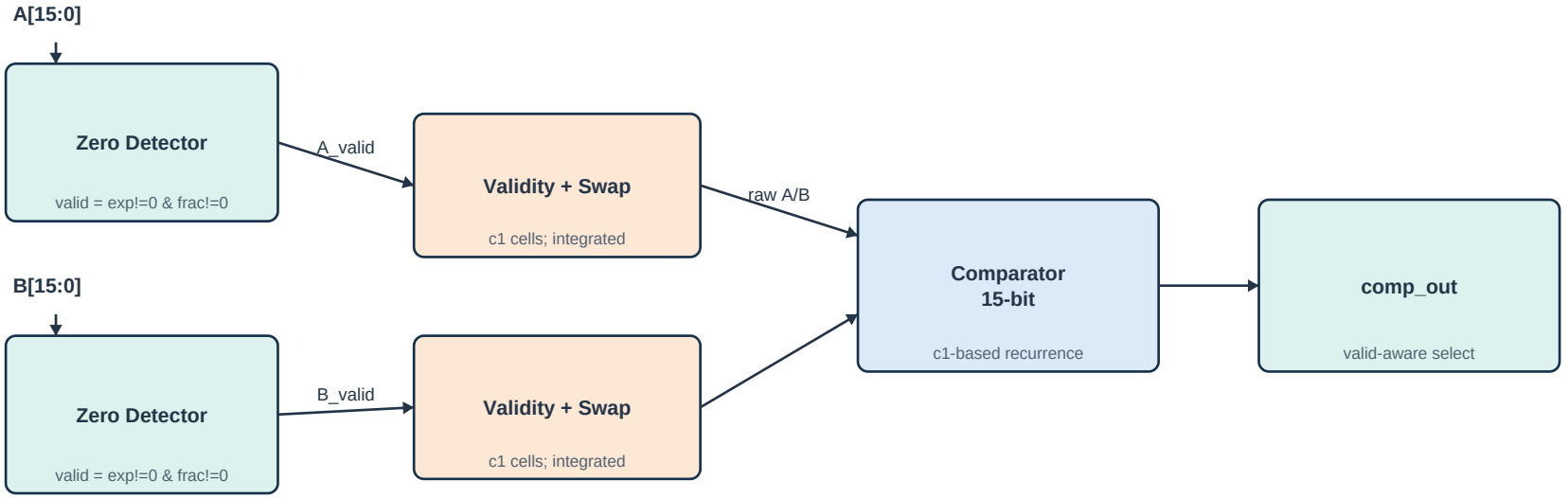
Component Schematics — Input Selection & Comparison



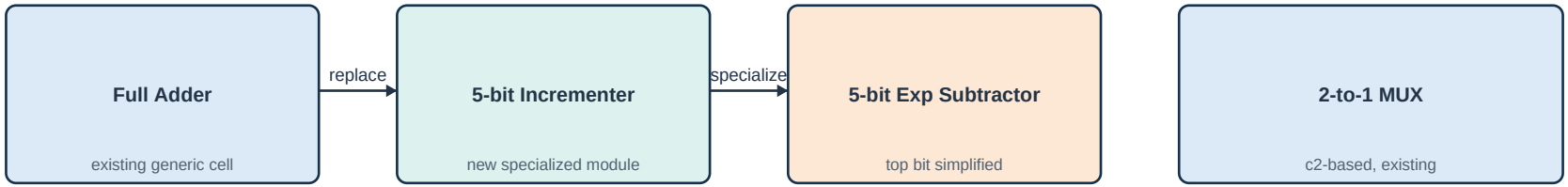
Optimization

- Removed separate 16-bit a_clean and b_clean muxes.
- Input validity is folded into operand selection.
- Comparator stages use c1 rather than c2.

Behavioral assumption

A valid operand has nonzero [14:0] under the project's existing validity rule. Verify with the original testbench before relying on equivalence.

Component Schematics — Arithmetic & Shifters



Increment_5_bit

Uses XOR stages for sum and AND stages for carry.
Avoids five general full-adder cells for +1.

Sub_exp_pe_5_bit

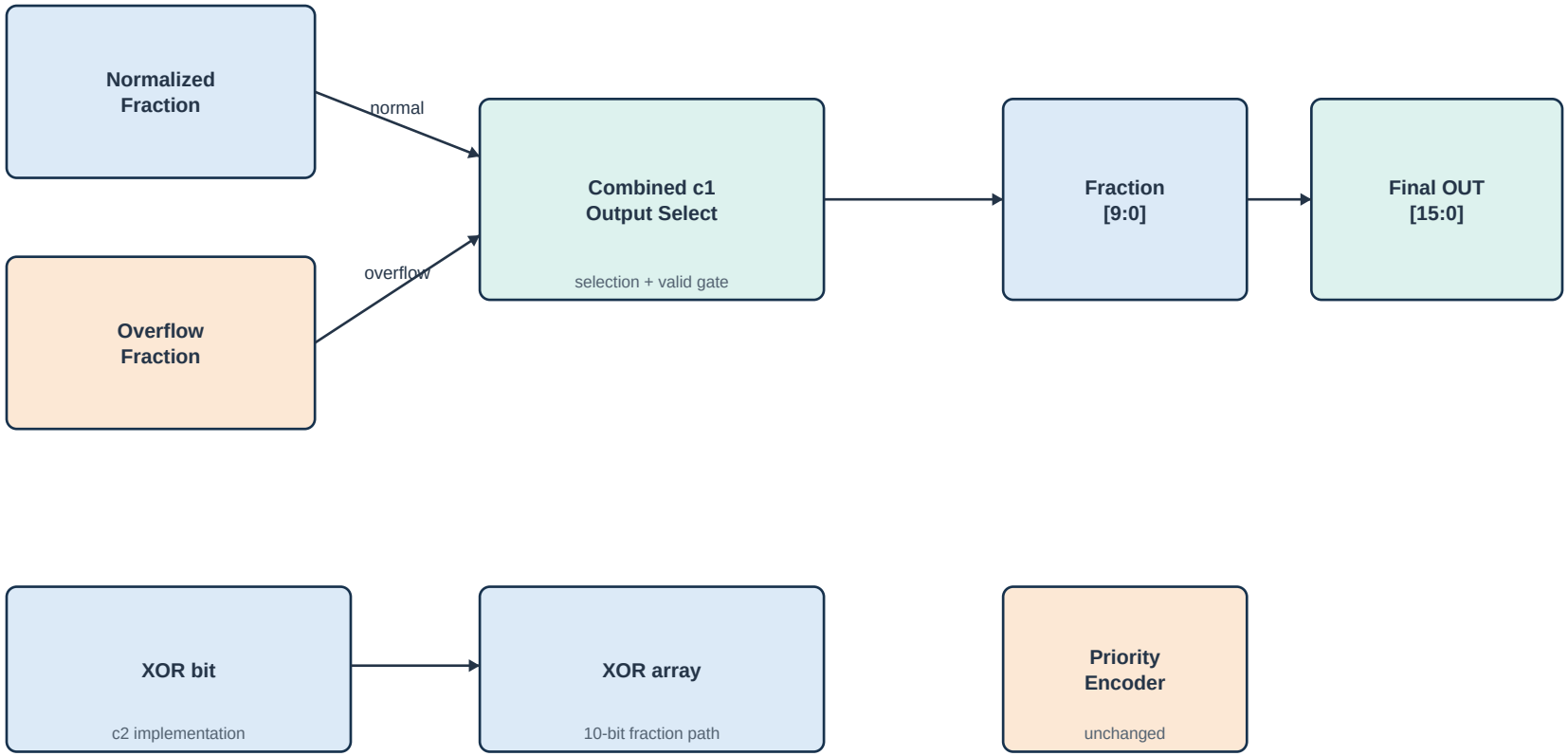
Second operand has a constant zero MSB.
The MSB carry/sum logic is therefore simplified.

Shift path

Width optimization

The left-shift result is only consumed as [9:0], so the optimized design uses Shift_left_10_bit.
This removes four mux cells (estimated saving: 40 gates).

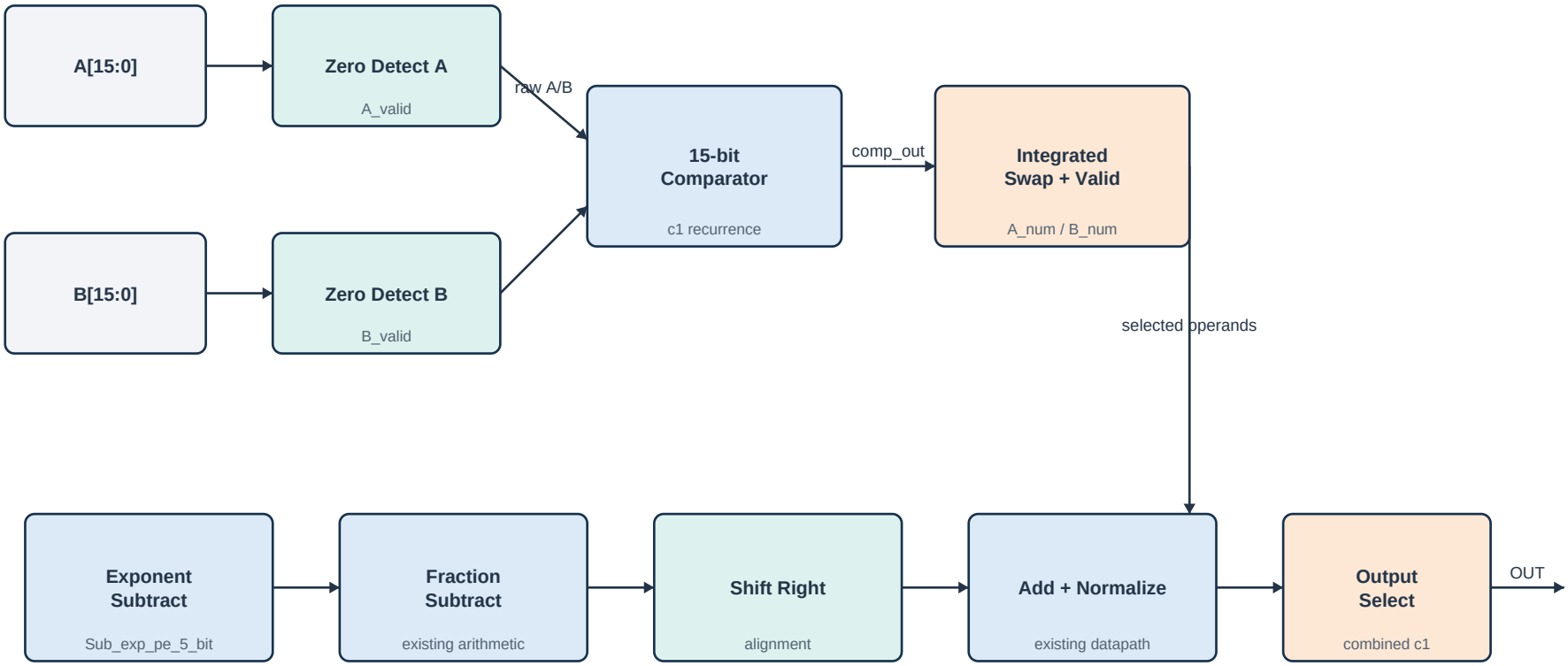
Component Schematics — Output Selection & Bitwise Logic



Output-path optimization

The fraction and exponent selection logic is combined with output-valid gating.
Estimated reduction: 15 c1 cells instead of separate branch muxes plus output gating.

Optimized FP16 Adder — Datapath Overview



Key structural changes

- Integrated input valid/swap path
- 10-bit left-shift result
- Specialized incrementer and exponent subtractor

Estimated gate impact

Estimated total reduction: 587 gates (3509 → 2922).
Estimate only; simulation and gated.txt must confirm.